

INCREMENTAL CONDITION ESTIMATION*

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Abstract. This paper introduces a new technique for estimating the smallest singular value, and hence the condition number, of a dense triangular matrix as it is generated one row or column at a time. It is also shown how this condition estimator can be interpreted as trying to approximate the secular equation with a simpler rational function. While one can construct examples where this estimator fails, numerical experiments demonstrate that despite its small computational cost, it produces reliable estimates. Also given is an example that shows the advantage of incorporating the incremental condition estimation strategy into the QR factorization algorithm with column pivoting to guard against near rank deficiency going unnoticed.

1. Introduction. Let $A = [a_1, \dots, a_n]$ be an $m \times n$ matrix and let $\sigma_1 \geq \dots \geq \sigma_{\min(m,n)} \geq 0$ be the singular values of A . The smallest singular value

$$\sigma_{\min} \equiv \sigma_{\min(m,n)}$$

of A is important in that it measures how close A is to a rank-deficient matrix [10, p. 19]. The condition number

$$\kappa_2(A) \equiv \frac{\sigma_1}{\sigma_{\min}}$$

which determines the sensitivity of equation systems involving A [10], [19], also depends crucially on $\sigma_{\min}$. For most practical purposes an order-of-magnitude estimate of $\sigma_{\min}$ or $\kappa_2(A)$ is sufficient. Most of the schemes for estimating $\sigma_{\min}$ and $\kappa_2(A)$ apply to triangular matrices, since in common applications A will be factored into a product of matrices involving a triangular matrix. An excellent survey of those so-called condition estimation techniques for triangular matrices as well as their applications is given by Higham [12].

All of these condition estimators do, however, estimate the smallest singular value of a triangular matrix *after* it has been factored and cannot be used to monitor the factorization of an upper triangular matrix as it is generated one column at a time (or a lower triangular matrix as it is generated one row at a time). Since a matrix and its transpose have the same singular values, we assume without loss of generality that we are generating a lower triangular matrix L one row at a time. The advantage of the incremental condition estimator we present in this paper is that it does allow us to *update estimates of $\sigma_{\min}(L)$ and $\kappa_2(L)$ cheaply as L is generated* one row at a time. In particular, if we are given an $n \times n$ lower triangular matrix L , an approximate singular vector x such that $\sigma_{\min}(L) \approx 1/\|x\|_2$, and a new row (v^T, γ) of L , we are able to obtain a new approximate singular vector of

$$L' \equiv \begin{pmatrix} L & 0 \\ v^T & \gamma \end{pmatrix}$$

* Received by the editors January 6, 1989; accepted for publication September 6, 1989. This work was partially supported by the U.S. Army Research Office through the Mathematical Science Institute of Cornell University, by the Office of Naval Research under contract N00014-83-K-0640, by National Science Foundation under contract CCR 86-02310, and the Applied Mathematical Sciences subprogram of the Office of Energy Research, U. S. Department of Energy under contract W-31-109-Eng-38. Computations were performed in part at the facilities of the Cornell Computational Optimization Project, which is supported by the National Science Foundation under contract DMS-87-06133.

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such that $\sigma_{\min}(L') \approx 1/\|y\|_2$ with $3n$ flops¹ and *without accessing L again*. In this fashion, incremental condition estimation makes it possible to monitor the condition number of L as it is generated.

In the next section we motivate the idea behind incremental condition estimation and describe the algorithm. Section 3 explains why incremental condition estimation works and explores its limitations. Section 4 presents numerical results showing the robustness of the proposed scheme. In § 5 we give an example of the usefulness of incremental condition estimation in the context of the QR factorization with column pivoting. Lastly, we summarize our contributions and outline directions for further research.

2. Estimating the smallest singular value of a triangular matrix. A common idea underlying condition estimators [7], [8], [11] is to exploit the implication

$$Lx = d \implies \frac{1}{\sigma_{\min}(L)} = \|L^{-1}\|_2 \geq \frac{\|L^{-1}d\|_2}{\|d\|_2} = \frac{\|x\|_2}{\|d\|_2}$$

by generating a large norm solution x to a moderately sized right-hand side d and then to use

$$\hat{\sigma}_{\min}(L) := \frac{\|d\|_2}{\|x\|_2}$$

as an estimate for $\sigma_{\min}(L)$. We hope that x will be an approximate singular vector corresponding to the smallest singular value and that as a consequence $\hat{\sigma}_{\min}(L)$ will not be too much of an overestimate of $\sigma_{\min}(L)$.

For our incremental condition estimator we want to monitor $\sigma_{\min}(L)$ as L is generated one row at a time. As a consequence, it is not feasible to reaccess L , since that would require $O(n^2)$ flops at every updating step, which is too expensive. To be more precise, given a good estimate $\hat{\sigma}_{\min}(L)$ defined by a large norm solution x to $Lx = d$ and a new row (v^T, γ) of L , we want to obtain a large norm solution y to

$$L'y = \begin{pmatrix} L & 0 \\ v^T & \gamma \end{pmatrix} y = d'$$

without accessing L again. None of the condition estimators surveyed by Higham [12] has that property.

We achieve this objective by choosing the new right-hand side d' in a fashion that allows us to reuse the previous approximate singular vector x . The idea is as follows.

Given x such that $Lx = d$ with $\|d\|_2 = 1$ and $\sigma_{\min}(L) \approx 1/\|x\|_2$, find $s := \sin \varphi$ and $c := \cos \varphi$ such that $\|y\|_2$ is maximized where y solves

$$(1) \quad \begin{pmatrix} L & 0 \\ v^T & \gamma \end{pmatrix} y = \begin{pmatrix} sx \\ c \end{pmatrix}.$$

By setting up the problem in this way we obtain immediately

$$(2) \quad y = \begin{pmatrix} sx \\ \frac{c-s\alpha}{\gamma} \end{pmatrix}$$

¹ Either an addition or a multiplication of two floating-point numbers is counted as a *flop*.

where

$$\alpha = v^T x$$

and $\|d'\|_2 = \|d\|_2 = 1$. The proper (s, c) is found using (2) and expressing

$$(3) \quad \|y\|_2^2 = \frac{1}{\gamma^2} \begin{pmatrix} s \\ c \end{pmatrix} B \begin{pmatrix} s \\ c \end{pmatrix}$$

where

$$(4) \quad B = \begin{pmatrix} 1 + \beta & -\alpha \\ -\alpha & 1 \end{pmatrix}$$

with

$$(5) \quad \beta = \gamma^2 x^T x + \alpha^2 - 1.$$

Assuming that $\gamma \neq 0$ (otherwise L' is singular and $\sigma_{\min}(L') = 0$), the optimal $\begin{pmatrix} s \\ c \end{pmatrix}$ is the eigenvector corresponding to the largest eigenvalue $\lambda_{\max}$ of B . In the case $\alpha \neq 0$ we define

$$(6) \quad \eta = \frac{\beta}{2\alpha} \quad \text{and} \quad \mu = \eta + \text{sign}(\alpha)\sqrt{\eta^2 + 1}$$

and obtain

$$(7) \quad \lambda_{\max} = \alpha\mu + 1$$

and

$$(8) \quad \begin{pmatrix} s \\ c \end{pmatrix} = \frac{1}{\sqrt{\mu^2 + 1}} \begin{pmatrix} \mu \\ -1 \end{pmatrix}.$$

For the special case $\alpha = 0$ we obtain

$$B = \begin{pmatrix} \gamma^2 x^T x & 0 \\ 0 & 1 \end{pmatrix}$$

in (4) and choose

$$\begin{pmatrix} s \\ c \end{pmatrix} = \begin{cases} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \text{if } |\gamma| \|x\|_2 > 1, \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{otherwise.} \end{cases}$$

Having computed the optimal $\begin{pmatrix} s \\ c \end{pmatrix}$, we compute a new approximate singular vector y as defined by (2) and the resulting estimate for the smallest singular value of $\sigma_{\min}(L')$ of L' is

$$(9) \quad \hat{\sigma}_{\min}(L') = \frac{1}{\|y\|_2}.$$

Given L , we need save only the current approximate singular vector x to compute an estimate for the smallest singular value of L' . Furthermore, the calculation is inexpensive. Since

$$(10) \quad \|y\|_2 = \frac{\sqrt{\lambda_{\max}}}{|\gamma|}$$

we need only $3k$ flops (a dot product and a scaling) to arrive at an estimate for $\sigma_{\min}(L')$. In particular, it costs only $\frac{3}{2}n^2$ flops to run this condition estimator alongside the generation of an $n \times n$ triangular matrix.

The incremental condition estimation idea can also be used to obtain good estimates for $\sigma_{\max}$, the largest singular value. Traditionally, the norm of the largest column, i.e.,

$$(11) \quad r_{\max} := \max_{1 \leq i \leq n} \|a_i\|_2 \quad (\leq \sigma_{\max})$$

is used as an estimate for $\sigma_{\max}(A)$. It is easy to show that

$$r_{\max} \sqrt{n} \geq \sigma_{\max}$$

and hence $r_{\max}$ underestimates $\sigma_{\max}$ at most by a factor of $\sqrt{n}$. For large n this could be a substantial underestimate. Furthermore, in the estimate

$$(12) \quad \hat{\kappa}_2(L) = \frac{r_{\max}}{\hat{\sigma}_{\min}} \quad (\leq \kappa_2(L)),$$

the errors in both $r_{\max}$ and $\hat{\sigma}_{\min}$ multiply.

To obtain a better estimate for $\sigma_{\max}$, observe that an approximate singular vector x for $\sigma_{\max}$ is a *small norm solution* to $Lx = d$, $\|d\|_2 = 1$ for a suitably chosen vector d . Note that this vector d is different from the one involved in estimating the smallest singular value. Given such a vector x for $Lx = d$, we compute (s, c) such that $\|y\|_2$ is *minimized* where y solves (1). The optimal (s, c) is then computed as in (5) through (8), except that we define

$$\mu = \eta - \text{sign}(\alpha) \sqrt{\eta^2 + 1}.$$

We mention that the incremental condition estimation scheme is related to the two-norm condition estimator suggested by Cline, Conn, and Van Loan [7], [18]. While their scheme looks backward and forward in a matrix, our scheme looks only backward to allow for the estimator to proceed in an incremental fashion.

3. Limitations. Given that there are counterexamples for computationally more expensive condition estimation schemes [12], it is not surprising that incremental condition estimation can fail as well. To understand why incremental condition estimation works, consider the singular value decomposition

$$L = U \Sigma V^T, U = [u_1, \dots, u_n], V = [v_1, \dots, v_n]$$

of L and assume that incremental condition estimation was exact and computed

$$(13) \quad x = \frac{1}{\sigma_{\min}(L)} v_n.$$

x is the largest possible solution to an equation system $Lx = d$, where $\|d\|_2 = 1$. Letting

$$L' = \begin{pmatrix} L & 0 \\ v^T & \gamma \end{pmatrix}$$

consider the so-called *secular equations* [4], [5] for the singular values of L' . The roots

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{n+1} \geq 0$$

of

$$(14) \quad f(\lambda) := \sum_{i=1}^n \frac{(v^T v_i)^2}{\sigma_i^2(L) - \lambda} - \frac{\gamma^2}{\lambda} + 1$$

determine the singular values of L' in that

$$\sigma_i^2(L') = \lambda_i.$$

Now define

$$(15) \quad \tilde{f}(\lambda) := \frac{(v^T v_n)^2}{\sigma_{\min}^2(L) - \lambda} - \frac{\gamma^2}{\lambda} + 1.$$

The smallest root $\tilde{\lambda}$ of $\tilde{f}$ is

$$\tilde{\lambda} = \frac{1}{2} (\tau - \sqrt{\tau^2 - 4\gamma^2 \sigma_{\min}^2(L)})$$

where

$$\tau = \sigma_{\min}^2(L) + (v^T v_n)^2 + \gamma^2.$$

f and $\tilde{f}$ are identical if

$$|v^T v_n| = \|v\|_2$$

and then

$$\tilde{\lambda} = \lambda_{n+1} = \sigma_{\min}^2(L').$$

In this special case it is also easy to verify that $\lambda_{\max}$, as defined in (7), is

$$\lambda_{\max} = \frac{1}{2\sigma_{\min}^2(L)} (\tau + \sqrt{\tau^2 - 4\gamma^2 \sigma_{\min}^2(L)})$$

and the resulting estimate for $\hat{\sigma}_{\min}(L')$ defined by (9) and (10) satisfies

$$\hat{\sigma}_{\min}^2(L') = \sqrt{\tilde{\lambda}}$$

and hence is exact.

This analysis shows that incremental condition estimation can be viewed as trying to approximate the secular equation (14) with the simpler rational function (15). This is a reasonable strategy, since the smallest root $\tilde{\lambda}$ of $\tilde{f}$ is a good approximation to the

smallest root λ_{n+1} of f when the roots of f are reasonably well separated and $v^T v_n$ is not too small in comparison with $\|v\|_2$.

In the general case, $\tilde{\lambda} \neq \sigma_{\min}(L')^2$, but instead $\tilde{\lambda} = \sigma_{\min}(\tilde{L})^2$, where

$$\tilde{L} = \begin{pmatrix} L & 0 \\ \tilde{v} & \gamma \end{pmatrix}$$

with

$$\tilde{v} := \frac{v^T x}{\|x\|_2^2} x = (v^T v_n) v_n$$

being the projection of v onto $\text{span}(v_n) = \text{span}(x)$. Hence, incremental condition estimation approximates $\sigma_{\min}(L')$ by $\sigma_{\min}(\tilde{L})$.

This analysis shows that incremental condition estimation can be unreliable when

$$(16) \quad |v^T x| \ll \|x\|_2 \|v\|_2.$$

Then the singular values of L' and $\tilde{L}$ can differ by an arbitrary amount. It should be noted, however, that even in the case where v and x are orthogonal, incremental condition estimation need not necessarily fail. If γ is small enough, the contribution of v to the change of the smallest singular values is negligible and incremental condition estimation still works. This is the reason why in the QR factorization with column pivoting [6], [9], [10] the smallest diagonal element usually works as a predictor of $\sigma_{\min}$.

We also point out that the previous analysis leads to a counterexample for the two-norm condition estimator suggested by Cline, Conn, and Van Loan [7], [18], since the n th step of their condition estimator is identical to incremental condition estimation. No counterexample for this condition estimator had been known thus far.

4. Numerical experiments. To assess the reliability of incremental condition estimation, we performed the test suite suggested by Higham [12] using PRO-MAT-LAB [16]. Three different types of test matrices are employed. In each test, upper triangular matrices R were generated by computing the Householder QR factorization of various $n \times n$ matrices A for $n = 25, 50, 75, 100$ both with and without column pivoting.

Test 1. The elements of A were chosen as random numbers from the uniform distribution on $[-1, 1]$. Fifty matrices were generated for each n . As observed by Higham, this type of matrix usually is well conditioned. Over the whole test the minimum, maximum, and average values of the two-norm condition number $\kappa_2(A) = \sigma_1/\sigma_n$ were $24, 1.5 \cdot 10^4$, and $2.4 \cdot 10^6$, respectively.

Tests 2 and 3. In these tests we used random matrices A with preassigned singular value distributions $\{\sigma_i\}$. Random orthogonal matrices U and V were generated using the method of Stewart [17] and then A was formed as $A = U\Sigma V^T$. For each value of n and each singular value distribution, fifty matrices were generated by choosing different matrices U and V . For Test 2 we chose the exponential distribution

$$\sigma_i = \alpha^i, \quad 1 \leq i \leq n$$

where $\alpha < 1$ is determined by $\kappa_2(A)$. For Test 3, we chose the sharp-break distribution

$$1 = \sigma_1 = \cdots = \sigma_{n-1} > \sigma_n = \frac{1}{\kappa_2(A)}.$$

TABLE 1
Max/avg values of $\hat{\sigma}_{\min}(R)/\sigma_{\min}(R)$.

Test 1: Uniform Distribution of Singular Values		
n	No pivoting	Column pivoting
25	6.6/2.4	2.9/2.0
50	7.0/3.1	5.6/2.6
75	8.5/3.5	4.9/3.2
100	9.6/4.1	5.5/3.4

Test 2: Exponential Distribution of Singular Values						
	No pivoting			Column pivoting		
n	$\kappa_2 = 10$	$\kappa_2 = 10^6$	$\kappa_2 = 10^{12}$	$\kappa_2 = 10$	$\kappa_2 = 10^6$	$\kappa_2 = 10^{12}$
25	1.6/1.3	4.2/2.5	11/3.6	1.7/1.4	3.3/2.2	4.1/2.1
50	1.7/1.4	6.1/2.9	6.9/3.8	1.6/1.4	3.4/2.4	4.1/3.0
75	1.5/1.4	4.0/3.0	6.1/3.9	1.6/1.4	3.5/2.7	5.1/3.1
100	1.5/1.4	4.3/3.0	7.3/4.1	1.6/1.4	3.4/2.8	5.0/3.5

TABLE 2
Max/avg values of $\kappa_2(R)/\hat{\kappa}_2(R)$.

Test 1: Uniform Distribution of Singular Values		
n	No pivoting	Column pivoting
25	6.8/2.4	2.9/2.0
50	7.0/3.1	5.6/2.6
75	8.6/3.5	4.9/3.2
100	9.7/4.1	5.5/3.4

Test 2: Exponential Distribution of Singular Values						
	No pivoting			Column pivoting		
n	$\kappa_2 = 10$	$\kappa_2 = 10^6$	$\kappa_2 = 10^{12}$	$\kappa_2 = 10$	$\kappa_2 = 10^6$	$\kappa_2 = 10^{12}$
25	2.2/1.6	6.8/3.0	21/4.1	2.1/1.5	3.3/2.2	4.1/2.1
50	2.1/1.6	7.1/3.5	9.7/4.5	1.8/1.6	3.5/2.6	4.2/3.0
75	1.8/1.6	7.0/3.6	9.7/4.8	1.8/1.6	4.4/2.9	5.9/3.3
100	1.8/1.6	5.2/3.8	8.2/5.1	1.9/1.6	3.9/3.1	5.4/3.7

The figures given in Table 1 are the ratios

$$\hat{\sigma}_{\min}(R)/\sigma_{\min}(R) \geq 1.$$

The figures in Table 2 are the ratios

$$\kappa_2(R)/\hat{\kappa}_2(R) \geq 1$$

where

$$\hat{\kappa}_2(R) = \frac{\hat{\sigma}_{\max}}{\hat{\sigma}_{\min}}$$

and both $\hat{\sigma}_{\max}$ and $\hat{\sigma}_{\min}$ have been computed using incremental condition estimation. The first number in each pair is the maximum ratio over the fifty matrices, and the second is the average ratio. All results were rounded to two significant digits. For Test 3 we observed a ratio of 1.0 (i.e., the estimate had at least two correct figures) in all cases. These results show that our condition estimator is reliable in producing good estimates. We overestimate $\sigma_{\min}(R)$ (or underestimate $\kappa_2(R)$) only by a small factor and the results vary only little with condition number, matrix size, and singular value distribution. Since

$$\frac{\kappa_2}{\hat{\kappa}_2} = \left(\frac{\hat{\sigma}_{\min}}{\sigma_{\min}} \right) \left(\frac{\sigma_{\max}}{\hat{\sigma}_{\max}} \right),$$

the comparison of Tables 1 and 2 shows that

$$\frac{\sigma_{\max}}{\hat{\sigma}_{\max}} \approx 1$$

in most cases, and hence the estimate for the largest singular value is very good.

We also note that although pivoting somewhat increases the accuracy of the condition estimator, it is not needed to obtain reliable estimates. This is in contrast to the estimator for $\sigma_{\min}$ arrived at choosing

$$\hat{\sigma}_{\min}(L) = \min_{1 \leq i \leq n} |l_{ii}|.$$

This estimator also works in an incremental fashion, but is unreliable when the matrix is not graded (as it is, for example, in the QR factorization with column pivoting). In particular, this condition estimator fails on matrices produced by the QR or Cholesky factorizations without column pivoting.

5. Guarding the QR factorization with column pivoting. A well-known strategy for extracting a set of reasonably independent columns of a given matrix A and for computing an orthonormal basis for the span of A is the QR factorization with column pivoting [1], [6], [15]. Viewed geometrically [10, p. 168, P.6.4-5] this strategy chooses at every step that column of A that is farthest away (in the two-norm sense) from the subspace spanned by the columns that were selected before. In matrix terms we are computing a QR factorization

$$(17) \quad AP = QR$$

where P is an $n \times n$ permutation matrix determined by the pivoting strategy, Q is an $m \times m$ orthogonal matrix, and R is an $m \times n$ upper triangular matrix. If A is dense, Q is typically generated as a sequence of Householder transformations [10, p. 37].

We hope that the rank of A will reveal itself by a small trailing subblock of R : if we partition R into

$$(18) \quad \begin{pmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{pmatrix}$$

with an $r \times r$ lower right-hand block R_{22} , then it is easy to show [10, p. 19] that

$$\sigma_{n-r+1}(A) \leq \|R_{22}\|_2.$$

Hence, if R_{22} is small, A can be considered to have numerical rank $n - r$, and the first $n - r$ columns of Q form an orthonormal basis for the range space of A .

This strategy works well in practice, but there are counterexamples where it fails without giving any indication of failure. A well-known example (originally suggested by Kahan) is

$$(19) \quad A_n = \text{diag}(1, s, s^2, \dots, s^{n-1}) \begin{pmatrix} 1 & -c & \cdots & \cdots & -c \\ 0 & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 1 & -c \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix} + \Delta$$

where $\Delta = \text{diag}(n\epsilon, (n-1)\epsilon, \dots, \epsilon)$, $c^2 + s^2 = 1$, and ϵ is the machine precision. A_n is very ill conditioned, but although each leading principal submatrix A_k ($k \leq n$) is also ill conditioned, there is a well-defined gap between σ_n and σ_{n-1} . As an example, for $n = 50$ and $c = 0.5$ we have $\sigma_{49} = 1.2 \cdot 10^{-3}$ and $\sigma_{50} = 3.7 \cdot 10^{-12}$. Even in floating-point arithmetic the matrix is its own QR factorization with pivoting but no trailing block of R is small enough to reveal its ill conditioning.

However, the incremental condition estimator integrated into the QR factorization algorithm detects the ill conditioning of the leading principal submatrices A_k and, in fact, we observed that in this particular example it never overestimates the smallest singular value of A_k , $k = 1, \dots, 50$ by a factor of more than 1.5. So while the column pivoting scheme fails to detect rank deficiency, the incremental condition estimator prevents this failure from going unnoticed. Given its negligible cost compared to the QR factorization, this suggests the usefulness of incorporating the incremental condition estimator into the traditional column pivoting scheme. In the same spirit we believe incremental condition estimation to be useful in monitoring Gaussian elimination and Cholesky factorization. The traditional “after-the-fact” condition estimation schemes, on the other hand, would only indicate that there had been problems at some unspecified point in the factorization process.

6. Conclusions. We introduced a technique that allowed us to estimate the smallest singular value of a dense triangular matrix R as it was generated one row (or column) at a time. This strategy required only $O(n)$ flops per step and the storage of $O(n)$ words between successive steps. In particular, it was not necessary to reaccess the previously generated R when a new row or column was added to R . We also showed how this strategy is related to the approximation of the secular equations. While one can construct examples where this strategy fails, numerical experiments indicate that the suggested scheme is reliable despite its small computational cost.

In the context of the QR factorization with column pivoting we gave an example that showed the usefulness of integrating the incremental condition estimator within the factorization process. Another use of incremental condition estimation is finding those columns of a matrix that are responsible for its ill conditioning. The Householder QR factorization algorithm with column pivoting usually achieves this goal, but the traditional pivoting strategy may conflict with other desirable features such as the sparsity structure of a matrix or locality of memory reference in the program.

Incremental condition estimation performs well on all kinds of triangular matrices, whereas the (also incremental) estimate of taking the smallest diagonal entry works only on the graded matrices produced by the traditional pivoting strategy. Incremental condition estimation allows us to restrict pivoting without sacrificing numerical reliability, and hence one can tailor the pivoting strategy to conform to other requirements. An example of such an application in the context of a rank-identifying QR factorization on a distributed-memory MIMD machine is given in [3]. Other applications we are currently exploring are block QR factorization schemes for rank-deficient matrices and algorithms for rank-deficient sparse matrices.

We also mention that it would be preferable to also have a *lower* bound for $\sigma_{\min}(R)$ instead of the upper bound that the incremental condition estimator is computing. To that end we experimented with the lower bounds derived from comparison matrices [2], [13], [14] that can also be updated in an incremental fashion. We found, however, that these bounds in most cases underestimated the smallest singular value by several orders of magnitude (this is consistent with Higham's [12] results) and as a result were not of practical use.

REFERENCES

- [1] A. BJÖRCK, *Least squares methods*, in Handbook of Numerical Analysis, Vol. 1, P. Ciarlet and J. Lions, eds., North Holland, Amsterdam, 1989, to appear.
- [2] N. ANDERSON AND I. KARASALO, *On computing bounds for the least singular value of a triangular matrix*, BIT, 15 (1975), pp. 1–4.
- [3] C. H. BISCHOF, *A parallel QR factorization algorithm with controlled local pivoting*, Tech. Report ANL/MCS-P21-1088, Argonne National Laboratory, Mathematics and Computer Sciences Division, Argonne, IL, 1988.
- [4] J. R. BUNCH AND C. R. NIELSEN, *Updating the singular value decomposition*, Numer. Math., 31 (1978), pp. 111–129.
- [5] J. R. BUNCH, C. R. NIELSEN, AND D. C. SORENSEN, *Rank-one modification of the symmetric eigenproblem*, Numer. Math., 31 (1978), pp. 31–48.
- [6] P. A. BUSINGER AND G. H. GOLUB, *Linear least squares solution by Householder transformation*, Numer. Math., 7 (1965), pp. 269–276.
- [7] A. K. CLINE, A. R. CONN, AND C. F. VAN LOAN, *Generalizing the LINPACK Condition Estimator*, Lecture Notes in Mathematics 909, Springer-Verlag, Berlin, New York, 1982, pp. 73–83.
- [8] A. K. CLINE, C. B. MOLER, G. W. STEWART, AND J. H. WILKINSON, *An estimate for the condition number of a matrix*, SIAM J. Numer. Anal., 16 (1979), pp. 368–375.
- [9] G. H. GOLUB, *Numerical methods for solving linear least squares problems*, Numer. Math., 7 (1965), pp. 206–216.
- [10] G. H. GOLUB AND C. F. VAN LOAN, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, London, 1983.
- [11] N. J. HIGHAM, *Efficient algorithms for computing the condition number of a tridiagonal matrix*, SIAM J. Sci. Statist. Comput., 7 (1986), pp. 150–165.
- [12] ———, *A survey of condition number estimation for triangular matrices*, SIAM Rev., 29 (1987), pp. 575–596.
- [13] ———, *Upper bounds for the condition number of a triangular matrix*, Numerical Analysis Report No. 86, University of Manchester, England, 1983.

- [14] I. KARASALO, *A criterion for truncation of the QR-decomposition algorithm for the singular linear least squares problem*, BIT, 14 (1974), pp. 156–166.
- [15] C. L. LAWSON AND R. J. HANSON, *Solving Least Squares Problems*, Prentice-Hall, Englewood Cliffs, NJ, 1974.
- [16] C. MOLER, J. LITTLE, AND S. BANGERT, *PRO-MATLAB User's Guide*, The Mathworks, Sherborn, MA, 1987.
- [17] G. W. STEWART, *The efficient generation of random orthogonal matrices with an application to condition estimators*, SIAM J. Numer. Anal., 17 (1980), pp. 403–409.
- [18] C. F. VAN LOAN, *On estimating the condition of eigenvalues and eigenvectors*, Linear Algebra Appl., 88/89 (1987), pp. 715–732.
- [19] J. H. WILKINSON, *The Algebraic Eigenvalue Problem*, Clarendon Press, Oxford, 1965.

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